

1

Multiple choice 1 point Question

Comparing arrays and linked lists, which statement is true?

- ☐ Data in both linked lists and arrays is stored in contiguous (samliggjandi) memory locations
- ☐ Data in neither linked lists nor arrays is stored in contiguous (samliggjandi) memory locations
- ☒ Data in arrays is stored in contiguous (samliggjandi) memory locations but not linked lists
- ☐ Data in linked lists is stored in contiguous (samliggjandi) memory locations but not arrays

2

Multiple choice 1 point Question

The time complexity for fetching a value from a singly-linked list at a specified node (item number n in the list) is:

- ☐ $O(\log_2 n)$
- ☐ $O(1)$
- ☒ $O(n)$
- ☐ $O(n^2)$

3

Multiple choice 1 point Question

The time complexity for adding a new node at the front (head) of a singly-linked list is:

- ☐ $O(\log_2 n)$
- ☒ $O(1)$
- ☐ $O(n)$
- ☐ $O(n^2)$

4

Multiple choice 1 point Question

In order to access the next-to-last node (in front of tail) in a singly-linked list the most efficient possible way for a program is to:

- ☐ index directly to it using it's index in the list ($O(1)$)
- ☐ go from the tail (last node) to its previous node in one step ($O(1)$)
- ☐ jump from the head (first node), in steps of $n/2$ nodes at a time, where n is the number of nodes between the current location and the tail (last node) ($O(\log n)$)
- ☒ walk the entire list from the head (first node), one step at a time ($O(n)$)

5

Multiple choice 1 point Question

When a linked list does not have reserved memory for an item that is to be added, the program must:

- ☐ reserve data for several more items in a new memory location
- ☒ just add another item, as there is no capacity limit on a linked list
- ☐ re-allocate memory for the entire list and copy the data to the new location
- ☐ stop adding, as you can not add to a linked list after the capacity is used up